

ME 333
MECHANICS OF MATERIALS

Lecture 2: Generalized Strain Definition

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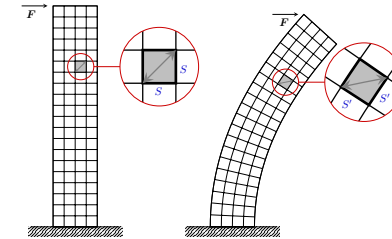
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Concept of Strain

Terminology - Review

Particle: A small amount of mass that occupies a single point in space (*of course, in relative terms*).

Rigid Body: A collection of many particles in which the distance between particles is fixed both before and after loading ($S = S'$).

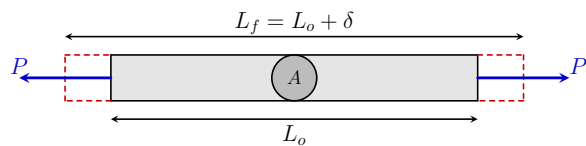


Deformable Body: If loading results in deformations (relative change in distance between particles) of the body ($S \neq S'$).

Concept of Strain

Strain in 1-D

Q Define Strain ?

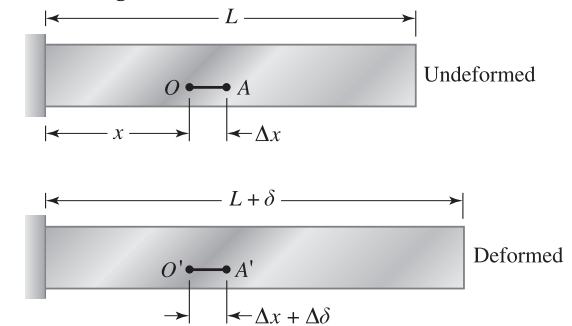


- $\epsilon = \frac{\text{Change in Length}}{\text{Original Length}} = \frac{L - L_0}{L_0} = \frac{\delta}{L_0}$
- δ is relative displacement and L_0 is the original length
- If the deformation is uniform along the length of the bar
- Deformation is the change in the relative position of points
- Strain is dimensionless and therefore no units

Concept of Strain

Average Strain Definition

[Example 1] Uniaxial Loading of a Bar



$$\epsilon_{\text{ave}} = \frac{(L + \delta) - L}{L} = \frac{\delta}{L} \quad (\text{Normal Strain})$$

Strain at a Point in 1-D

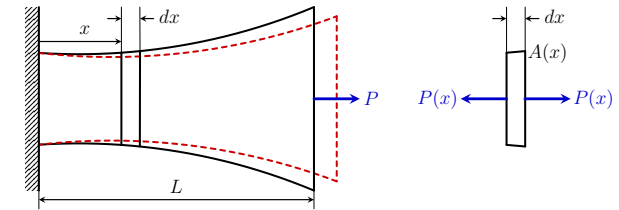
Q What if the deformation is not uniform ?

- If the deformation is not uniform:

$$\epsilon(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta \delta}{\Delta x} = \frac{d\delta}{dx} \quad \Rightarrow \quad d\delta = \epsilon(x) dx \quad \therefore \delta = \int_0^L \epsilon(x) dx$$

Strain at a Point in 1-D

Q What if the deformation is not uniform ? [Example 1]

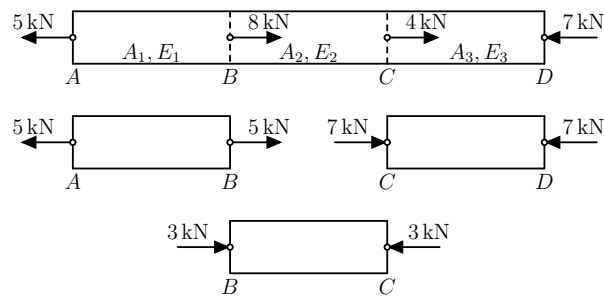


- If deformation is not uniform:

$$\epsilon(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta \delta}{\Delta x} = \frac{d\delta}{dx} \quad \Rightarrow \quad d\delta = \epsilon(x) dx \quad \therefore \delta = \int_0^L \epsilon(x) dx$$

Strain at a Point in 1-D

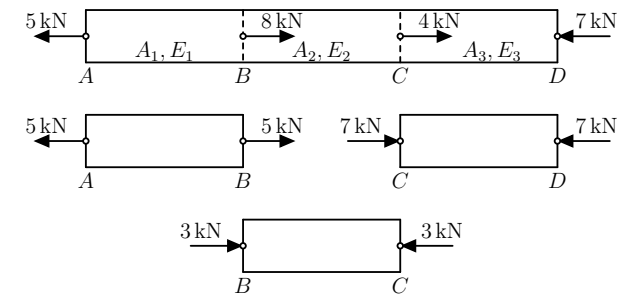
[Example 2] Bar AD is loaded as shown. Find the relative displacement δ_{AD} .



- Draw free body diagram of each section

Strain at a Point in 1-D

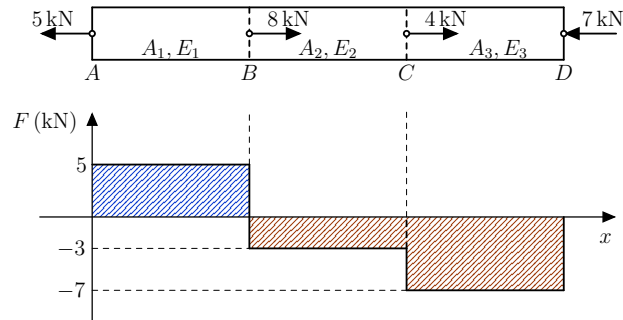
[Example 2] Contd ...



$$\delta_{AD} = \sum_{i=1}^N \left(\frac{P_i L_i}{A_i E_i} \right) = \left(\frac{5L_{AB}}{A_1 E_1} \right) - \left(\frac{3L_{BC}}{A_2 E_2} \right) - \left(\frac{7L_{CD}}{A_3 E_3} \right)$$

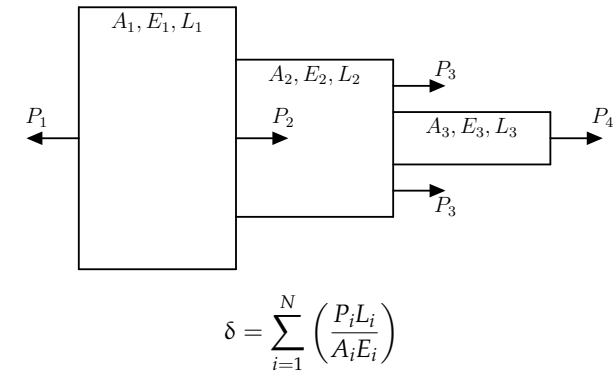
Strain at a Point in 1-D

Axial Force Diagram



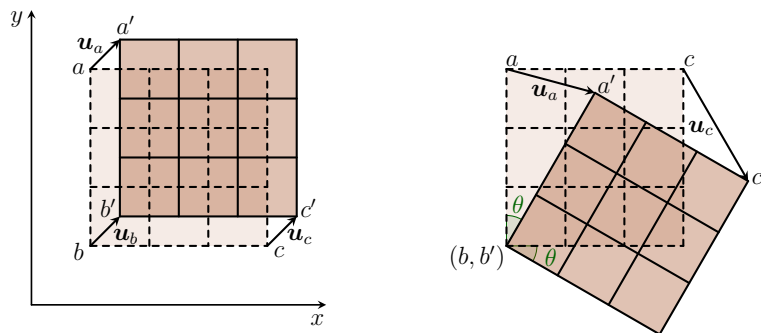
Strain at a Point in 1-D

[Example 3] Stepped bar under axial loading.



Types of Motion

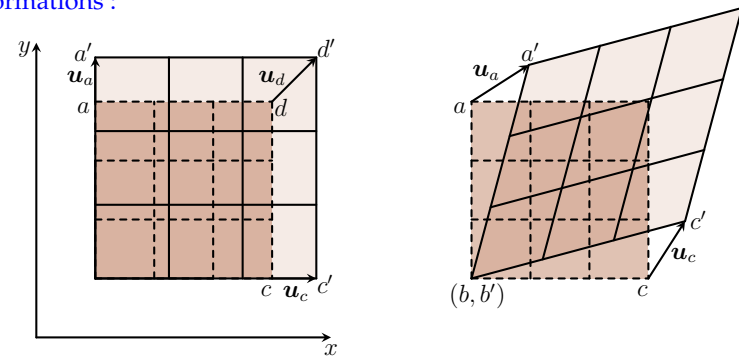
Rigid Body Motions :



- No relative change in distance between the points of the grid.

Types of Motion

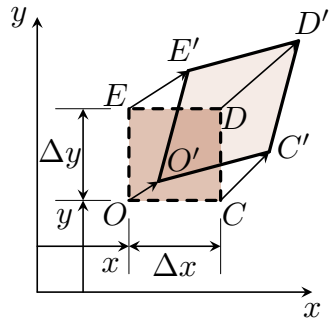
Deformations :



- Relative change in distance between the points of the grid (*Dilation, Distortion*).

Deformation of Infinitesimal Element

Deformations in xy plane :



- The strain components :

$$\epsilon_x = \lim_{\Delta x \rightarrow 0} \frac{O'C' - OC}{OC}$$

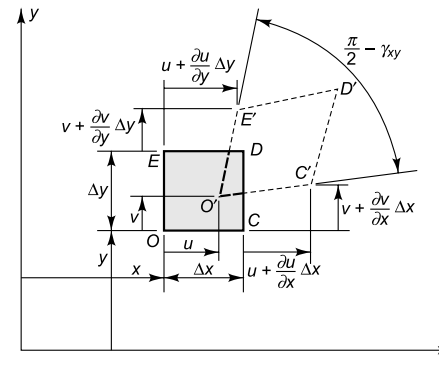
$$\epsilon_y = \lim_{\Delta y \rightarrow 0} \frac{O'E' - OE}{OE}$$

$$\begin{aligned} \gamma_{xy} &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} (\angle COE - \angle C'O'E') \\ &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \left(\frac{\pi}{2} - \angle C'O'E' \right) \end{aligned}$$

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Strain at a Point in 2-D

Deformations in xy plane :



- The strain components :

$$\epsilon_x = \lim_{\Delta x \rightarrow 0} \frac{\Delta x + (\partial u / \partial x) \Delta x - \Delta x}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\epsilon_y = \lim_{\Delta y \rightarrow 0} \frac{\Delta y + (\partial v / \partial y) \Delta y - \Delta y}{\Delta y} = \frac{\partial v}{\partial y}$$

$$\begin{aligned} \gamma_{xy} &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \left\{ \frac{(\partial v / \partial x) \Delta x}{\Delta x} + \frac{(\partial u / \partial y) \Delta y}{\Delta y} \right\} \\ &= \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \end{aligned}$$

Ref: An Introduction to Mechanics of Solids, Crandall et. al.

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Small Strain Theory

[Example 4] Rigid rotation about the z -axis:

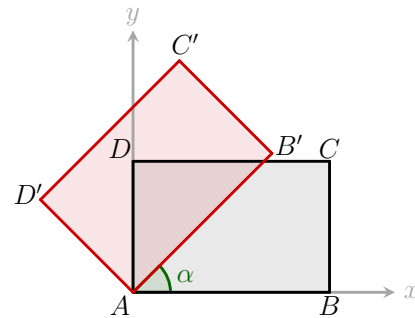
$$\begin{aligned} x' &= x + u; & y' &= y + v; & z' &= z + w \\ &= x \cos \alpha + y \sin \alpha; & &= -x \sin \alpha + y \cos \alpha; & &= z \end{aligned}$$

- Small strain theory

$$\epsilon_{xx} = \frac{\partial u}{\partial x} = \frac{\partial x'}{\partial x} - 1 = \cos \alpha - 1;$$

$$\epsilon_{yy} = \frac{\partial v}{\partial y} = \frac{\partial y'}{\partial y} - 1 = \cos \alpha - 1;$$

$$\begin{aligned} \gamma_{xy} &= \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \\ &= \frac{\partial x'}{\partial y} + \frac{\partial y'}{\partial x} = \sin \alpha - \sin \alpha = 0 \end{aligned}$$

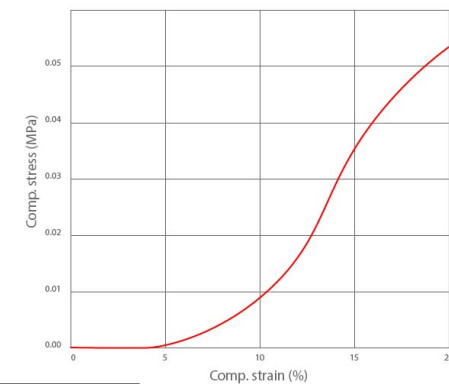


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Large Deformations

[Example 5] Deformations in a Tennis Ball

Compression Test

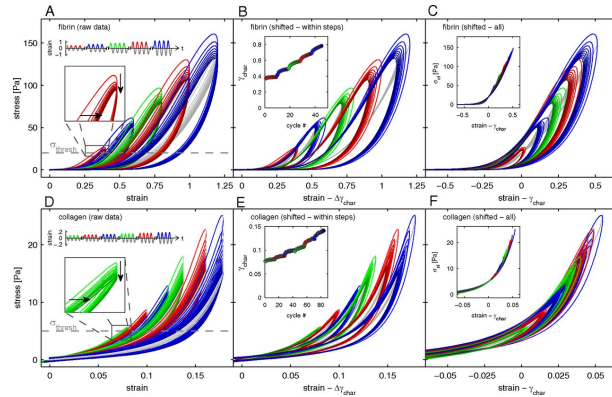


Ref: Instron Inc.

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Large Deformations

[Example 6] Deformations in biological materials



Ref: Stefan Münster et al., PNAS 2013

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17 / 21

Strain Measure

Q When is a body strained ?

- When relative position of points in a continuous body is altered, we say that the body is strained.
- **Rigid Body** : It is an ideal body such that the distance between every pair of points remains unaltered throughout history of body.
- Till now the definition of strain is one-dimensional. However, in reality strain is the *tensorial* measure in 3D.

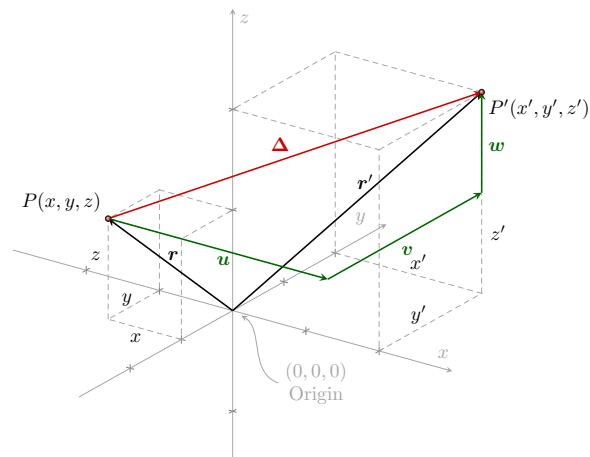
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18 / 21

Generalized Strain Description



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ME 333

Sem I, 2025-2026

19 / 21

Generalized Strain Description

Steps :

- 1 Consider a point $P(x, y, z)$ in an unloaded state (undeformed and unstressed)
- 2 Upon loading point P moves to $P'(x', y', z')$
- 3 The primed coordinates are functions of unprimed coordinates and vice-versa, i.e.,

$$\begin{aligned} x' &= x'(x, y, z); & x &= x(x', y', z') \\ y' &= y'(x, y, z); & y &= y(x', y', z') \\ z' &= z'(x, y, z); & z &= z(x', y', z') \end{aligned}$$

Requirement :

- 1 Above functions are continuous and differentiable
- 2 Above functions are invertible and single valued

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ME 333

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20 / 21

Reading Assignment

Chapter 2: Section 2.7.1